

THE UNIVERSITY OF ZAMBIA
SCHOOL OF NATURAL AND APPLIED SCIENCES
DEPARTMENT OF COMPUTING AND INFORMATICS

FINAL YEAR PROJECT REPORT

ACADEMIC YEAR 2026

[PROJECT TITLE]

A Final Year Project Report Submitted to the
Department of Computing and Informatics
in partial fulfilment of the requirements for the award of

[DEGREE PROGRAMME]

Student Name: [STUDENT NAME]

Computer Number: [COMPUTER NUMBER]

Supervisor: [SUPERVISOR NAME]

Date of Submission: [DATE]

DECLARATION

I declare that this final year project report is my own work and has been prepared in accordance with the academic requirements of the University of Zambia. All sources of information, ideas, data, software, images, and other material that are not my own have been appropriately acknowledged and referenced.

Where this project was undertaken as part of a group, I have clearly identified my individual contributions and have not presented the work of other group members as my own.

Student Name: _____

Computer Number: _____

Signature: _____

Date: _____

APPROVAL

This final year project report has been submitted for examination with the approval of the project supervisor.

Supervisor Name: _____

Signature: _____

Date: _____

Departmental/Examiner Approval (if applicable): _____

ABSTRACT

[Write the abstract here.]

Keywords: [keyword 1]; [keyword 2]; [keyword 3]; [keyword 4]; [keyword 5]

ACKNOWLEDGEMENTS

[Write acknowledgements here.]

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LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation	Meaning
API	[Application Programming Interface]
DBMS	[Database Management System]
ML	[Machine Learning]
UI	[User Interface]

NOTE ON GROUP PROJECTS

Students who worked on a project as a group are strongly encouraged to prepare their own individual final reports rather than submitting identical reports. The project, system, dataset, or overall research problem may be shared, but each student's report should clearly demonstrate that student's own understanding, responsibilities, work, results, and contribution to the project.

In particular, group-project students should:

- state the overall group project and explain how the individual work fits within it;
- clearly identify their own responsibilities and deliverables;
- focus detailed analysis, design, implementation, experiments, testing, and/or evaluation on their own contribution;
- acknowledge the work completed by other group members where it is discussed;
- include an *Individual Contribution Statement* in the appendices; and
- not claim another group member's work, results, code, data collection, or design decisions as their own.

Chapter 1

Introduction

1.1 Background and Context

Introduce the project area, context, motivation, and relevant background. Explain why the topic matters.

1.2 Problem Statement

Clearly define the specific problem addressed. Establish its relevance and justify why it requires investigation or a solution.

1.3 Project Aim

State one overall goal of the project.

1.4 Project Objectives

List specific, measurable objectives. Use the objectives established in the proposal, updating them if the project scope legitimately changed.

Objective 1

Objective 2

Objective 3

Objective 4

1.5 Research Questions (where applicable)

State the questions the research or investigation seeks to answer. Omit this section if it is not appropriate.

1.6 Scope of the Project

Define what is included and excluded.

1.6.1 In Scope

[Describe the elements included in the completed project.]

1.6.2 Out of Scope

[Describe important exclusions.]

1.7 Significance of the Project

Explain who benefits from the project and its practical, technical, research, or institutional significance.

1.8 Limitations

Discuss important constraints that affected the project, such as data, time, resources, access, or technical limitations.

1.9 Organisation of the Report

Briefly describe the contents of each subsequent chapter.

Chapter 2

Literature Review

2.1 Introduction

Introduce the purpose and structure of the literature review.

2.2 Key Concepts and Theories

Explain concepts, models, theories, frameworks, or technical foundations needed to understand the project.

2.3 Existing Systems / Related Work

Review relevant existing systems, studies, approaches, algorithms, technologies, or solutions.

2.4 Comparative Analysis

Compare relevant approaches or existing solutions using appropriate criteria.

2.5 Research / Knowledge Gap

Identify what is missing, inadequate, unresolved, or improvable in existing work.

2.6 Conceptual or Theoretical Framework (where applicable)

Present and explain the framework guiding the project.

2.7 Chapter Summary

Summarise the major conclusions from the literature and connect them to the project.

Chapter 3

Methodology

3.1 Introduction

Explain the methodology used to conduct the completed project.

3.2 Research / Development Approach

Describe and justify the approach used, such as Agile, Waterfall, experimental research, design science, or another appropriate method.

3.3 Requirements / Data Collection

Describe how requirements, data, participants, observations, or other inputs were obtained, where applicable.

3.4 System Architecture / Research Workflow

Present and explain the overall architecture, workflow, experimental pipeline, or research process.

3.5 Tools and Technologies

Document programming languages, frameworks/libraries, databases, hardware, platforms, development environments, and other major tools.

3.6 Ethical and Legal Considerations

Address data privacy, user consent, security, intellectual property, licensing, responsible use of data/software, and other relevant considerations.

3.7 Evaluation Method

Explain how success was measured, including test criteria, performance metrics, user evaluation, experimental design, or other evaluation procedures.

3.8 Chapter Summary

Summarise the methodology.

Chapter 4

System Analysis and Design

4.1 Introduction

Introduce the analysis and design activities.

4.2 Stakeholder and User Analysis

Identify relevant stakeholders and user groups.

4.3 Functional Requirements

Document what the system or solution must do.

4.4 Non-Functional Requirements

Document requirements such as performance, usability, security, reliability, maintainability, and scalability.

4.5 Use Cases / User Stories

Present appropriate use cases, user stories, or equivalent specifications.

4.6 System / Solution Architecture

Present the final architecture and explain its components and interactions.

4.7 Database / Data Design (where applicable)

Include ER diagrams, schemas, data models, or other relevant designs.

4.8 Interface / Interaction Design (where applicable)

Present relevant wireframes, interface designs, interaction flows, or screen designs.

4.9 Algorithms / Models / Technical Design

Explain important algorithms, models, technical decisions, or design choices.

4.10 Chapter Summary

Summarise the final design.

Chapter 5

Implementation / Development

5.1 Introduction

Introduce the implementation.

5.2 Development Environment

Describe the hardware and software environment used.

5.3 System Components / Modules

Describe the major components implemented.

5.4 Key Implementation Details

Explain important implementation decisions, algorithms, integrations, data processing, or model development.

5.5 Integration

Explain how components, services, APIs, databases, hardware, or models were integrated.

5.6 Challenges and Solutions

Discuss significant implementation challenges and how they were addressed.

5.7 Individual Contribution (group projects)

Clearly identify the student's own implementation work and distinguish it from work completed by other group members.

5.8 Chapter Summary

Summarise the implementation.

Chapter 6

Testing, Results and Evaluation

6.1 Introduction

Explain the purpose of testing and evaluation.

6.2 Testing Strategy

Describe testing levels and techniques used, such as unit, integration, system, usability, security, or acceptance testing.

6.3 Test Cases and Results

Present representative test cases, expected outcomes, actual outcomes, and pass/fail status.

Table 6.1: Example test-case structure

ID	Test Case	Expected Result	Status
TC-01	[Description]	[Expected result]	[Pass/Fail]
TC-02	[Description]	[Expected result]	[Pass/Fail]

6.4 Experimental Results / Performance Results (where applicable)

Present quantitative results, metrics, graphs, tables, or model evaluation.

6.5 User Evaluation (where applicable)

Present results from users, surveys, interviews, usability testing, or other evaluation.

6.6 Analysis of Results

Interpret the results rather than merely listing them.

6.7 Objective-by-Objective Evaluation

Explicitly show how the results demonstrate achievement of each project objective.

6.8 Individual Results (group projects)

Focus on results directly attributable to the student's contribution where the project was completed collaboratively.

6.9 Chapter Summary

Summarise the findings.

Chapter 7

Discussion

7.1 Introduction

Introduce the discussion.

7.2 Interpretation of Findings

Explain what the results mean in relation to the problem and objectives.

7.3 Comparison with Existing Work

Relate findings to the literature and existing solutions.

7.4 Contributions of the Project

Explain the project's practical, technical, research, or other contributions.

7.5 Limitations and Implications

Discuss limitations and their implications for interpreting the results.

7.6 Chapter Summary

Summarise the discussion.

Chapter 8

Conclusion and Recommendations

8.1 Introduction

Briefly introduce the conclusion.

8.2 Summary of the Project

Summarise the problem, approach, major work, and findings.

8.3 Conclusions

State the main conclusions supported by the project results.

8.4 Achievement of Objectives

State clearly whether and how each objective was achieved.

8.5 Recommendations

Provide practical recommendations arising from the project.

8.6 Future Work

Identify sensible extensions, improvements, or areas for further investigation.

REFERENCES

[Insert references here.] e.g.,

1. Chibuye, M., Phiri, J. (2025). Hybrid Prediction Model Using Elastic Net Regression and Echo State Networks for Enhanced Yield Prediction in Crop Systems: A Comparative Study of Dense Neural Networks and Hybrid Echo State Network Models. In: Proceedings of Tenth International Congress on Information and Communication Technology. ICICT 2025. Lecture Notes in Networks and Systems, vol 1412. Springer, Singapore. https://doi.org/10.1007/978-981-96-6429-0_29
2. Chibawe, G., Nyirenda, M. (2026). Enhancing Disease Outbreak Forecasting: Integrating Environmental Factors, Policy Interventions, and Machine Learning in Hybrid Epidemiological Models. In: ICT for Intelligent Systems. ICTIS 2025. Lecture Notes in Networks and Systems, vol 1511. Springer, Singapore. https://doi.org/10.1007/978-981-96-8796-1_13

Chapter A

Individual Contribution Statement (Group Projects)

Purpose of this Statement

This statement is required for students who completed their final year project as part of a group. Although the project may have been conceived, developed, and evaluated collaboratively, each student is expected to demonstrate their own understanding and contribution through an individual final report.

Students should therefore use this section to clearly identify the work for which they were primarily responsible. The statement should be consistent with the content presented in the main report.

Students should not claim the work of another group member as their own. Where a component, dataset, design, implementation, or result was produced collaboratively or by another group member, this should be clearly acknowledged.

A.1 Project and Group Information

Item	Details
Project Title	[Project Title]
Student Name	[Student Name]
Computer Number	[Computer Number]
Supervisor	[Supervisor Name]
Group Size	[Number of Students]
Group Members	[List all group members and computer numbers]

A.2 Overall Group Project

Provide a brief description of the overall project undertaken by the group. This should explain the common problem being addressed, the overall aim of the project, and the main system, application, model, research output, or solution developed by the group.

[Describe the overall group project here.]

A.3 Individual Role and Responsibilities

Describe the student's primary role within the group and the responsibilities assigned to them.

The description should identify the specific areas in which the student was expected to make a substantive contribution.

No.	Area of Responsibility	Description
1	[Area]	[Describe responsibility]
2	[Area]	[Describe responsibility]
3	[Area]	[Describe responsibility]
4	[Area]	[Describe responsibility]

A.4 Detailed Individual Contribution

Describe in detail the work personally undertaken by the student. The contribution may include, where applicable:

- requirements analysis;
- literature review and background research;
- system analysis;
- system or database design;
- user interface design;
- software development and implementation;
- algorithm development;
- machine learning model development;
- data collection, preparation, and analysis;
- hardware design or implementation;
- API or external-service integration;
- testing and debugging;

- performance evaluation;
- documentation;
- deployment and configuration; and
- other technical or research activities.

[Describe the student's individual contribution in detail.]

A.5 Individual Deliverables

List the specific outputs or deliverables produced primarily by the student.

No.	Deliverable	Student's Contribution
1	[Deliverable]	[Description]
2	[Deliverable]	[Description]
3	[Deliverable]	[Description]
4	[Deliverable]	[Description]

A.6 Contribution to the Main System or Research Output

Explain which components of the final system, research output, experiment, model, or solution were primarily developed or completed by the student.

Where a component was developed jointly with another group member, explicitly indicate the collaborative nature of the work.

[Describe contribution to the main project output here.]

A.7 Individual Implementation and Technical Work

Where the project involved software, hardware, data analysis, modelling, or other technical work, describe the student's specific technical contribution.

For software projects, students should identify the major modules, services, interfaces, algorithms, databases, integrations, or other components they implemented.

For research or data-oriented projects, students should identify the datasets, experiments, analyses, models, evaluation procedures, or other research activities for which they were responsible.

[Describe individual technical work here.]

A.8 Individual Testing and Evaluation Contribution

Describe the student's contribution to testing and evaluation of the project.

This may include test-case development, debugging, unit testing, integration testing, usability testing, performance evaluation, experimental analysis, model evaluation, or interpretation of results.

[Describe testing and evaluation contribution here.]

A.9 Collaboration with Other Group Members

Briefly explain how the student's work interacted with the work of other group members.

Identify important dependencies between components and acknowledge substantial contributions made by other members where relevant.

[Describe collaboration and dependencies here.]

A.10 Contribution Summary

Summarise the student's overall contribution to the project.

The summary should distinguish between:

1. work primarily completed by the student;
2. work completed jointly with other group members; and
3. work primarily completed by other group members but used or incorporated into the student's work.

[Provide contribution summary here.]

A.11 Estimated Contribution

Provide an approximate indication of the student's contribution across the major project activities.

Project Activity	Individual Contribution	Collaborative Contribution
Problem Analysis	[%]	[%]
Literature Review	[%]	[%]
Requirements Analysis	[%]	[%]
System/Research Design	[%]	[%]
Implementation / Development	[%]	[%]
Testing and Evaluation	[%]	[%]
Documentation	[%]	[%]
Other	[%]	[%]

Note: The percentages are intended to provide an approximate indication of the nature and distribution of the student's work. They are not necessarily expected to total 100% across rows.

A.12 Statement of Authorship and Contribution

I confirm that the contribution described in this statement accurately represents the work that I personally undertook as part of the group project. I have identified collaborative work and have not knowingly claimed the work of another group member as my own.

Student Name: _____

Computer Number: _____

Signature: _____

Date: _____

A.13 Supervisor Verification

I confirm that, to the best of my knowledge, the contribution described above is consistent with the student's participation in the group project.

Supervisor Name: _____

Signature: _____

Date: _____

Chapter B

User Manual / Installation Guide

[Insert supporting material.]

Chapter C

Additional System Designs / Diagrams

[Insert large diagrams, additional UML diagrams, database schemas, architecture diagrams, or other supporting design material.]

Chapter D

Additional Test Cases and Results

[Insert extended test cases, logs, screenshots, or detailed evaluation evidence.]

Chapter E

Data Collection Instruments

[Insert instruments where applicable.]

Chapter F

Additional Code / Configuration

[Insert selected supporting code/configuration.]

Chapter G

Other Supporting Material

[Insert any other material specifically required by the project or supervisor.]